

ENGR 213: Applied Ordinary Differential Equations

Quizzes & Term Tests Official Solved Examination Bank (Winter 2025)

Concordia University · Gina Cody School of Engineering and Computer Science

Official Document Bank: In-Class Quizzes 1–4 · Term Test 1 (Version 2) · Term Test 2 · 14 Comprehensive Solved Questions

Academic Session: Winter 2025 / Fall 2026 · **Pedagogical Standards:** Full Analytical Proofs & Exact Solutions

ENGR 213: Applied Ordinary Differential Equations

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Department of Mechanical, Industrial & Aerospace Engineering · Concordia University

Academic Session: Winter 2025 / Fall 2026 · **Course:** ENGR 213

Curriculum Mapping: In-Class Quizzes (Quizzes 1–4) · Term Test 1 (Version 2) · Term Test 2 · Comprehensive Step-by-Step Solutions

Executive Summary & Examination Blueprint

This master document compiles and rigorously solves all official in-class quizzes and departmental term tests administered in ENGR 213 during the Winter 2025 academic session. Every problem is numbered sequentially from **Question 1 through Question 14**, providing verbatim problem statements, classification theorems, step-by-step mathematical derivations, boxed final explicit answers, and professor grading insights.

Summary of Covered Examination Scope

- **Part I: Official In-Class Quizzes (Questions 1–4)**

* **Question 1 (Quiz #1-G · 15 Min):** First-Order Nonlinear Separable Initial Value Problem ($yy' + x^2 = 4, y(0) = 2$).

* **Question 2 (Quiz #2-G · 15 Min):** Physical Modeling of Rates of Change Inversely Proportional to the Square of the Dependent Variable ($dy/dt = k/y^2, y(0) = 1, y(1) = 2$).

* **Question 3 (Quiz #3-G · 15 Min):** Second-Order Constant-Coefficient Nonhomogeneous ODE via the Method of Undetermined Coefficients ($y'' + 5y' + 4y = \cos(3x)$).

* **Question 4 (Quiz #4 · 20 Min):** Power Series Solution about the Ordinary Point $x_0 = 0$ ($y'' - (x + 1)y = 0$, first 6 nonzero terms).

- **Part II: Official Term Test 1 — Version 2 (Questions 5–9 · Feb 21, 2025 · Marked on 100)**

* **Question 5 (Test 1 #1/20):** First-Order ODEs: (1) Direct Integration $\frac{dy}{dx} = \sin(3x)$ and (2) First-Order Linear ODE via Integrating Factor $y' + \frac{3}{x}y = x^4$.

* **Question 6 (Test 1 #2/15):** Complex Number Arithmetic: Addition, Division, Polar Modulus, and Multiplication ($z_1 = 4 + 5i, z_2 = -1 + 2i$).

* **Question 7 (Test 1 #3/20):** Exact Differential Equation IVP: $(5y + 3t - 5)dt + (6y + 5t)dy = 0, y(-1) = 0$.

* **Question 8 (Test 1 #4/25):** Logistic Differential Equation IVP for Technology Adoption:

$$\frac{dN}{dt} = N(1 - 0.0004N), N(0) = 2.$$

* **Question 9 (Test 1 #5/20):** Forensic Application of Newton's Law of Cooling: Time of Death and Ambient Thermal Decay.

• **Part III: Official Term Test 2 (Questions 10–14 · March 27, 2025 · Marked on 100)**

* **Question 10 (Test 2 #1/20):** Cauchy-Euler Equidimensional Second-Order ODE: $x^2y'' + 5xy' + 8y = 0$.

* **Question 11 (Test 2 #2/20):** Variation of Parameters for Repeated Root Second-Order ODE:

$$y'' - 4y' + 4y = (x + 1)e^{2x}.$$

* **Question 12 (Test 2 #3/20):** Second-Order Homogeneous Initial Value Problem with Critically Damped Roots:

$$y'' + y' + 0.25y = 0, y(0) = 3, y'(0) = -3.5.$$

* **Question 13 (Test 2 #4/20):** Mechanical Vibrations of a Damped Spring-Mass System (

$$m = 2 \text{ kg}, k = 72 \text{ N/m}, c = 4 \text{ N} \cdot \text{s/m}).$$

* **Question 14 (Test 2 #5/20):** Forced Damped Harmonic Oscillator IVP:

$$x'' + 4x' + 4x = 10 \cos(3t), x(0) = 0, x'(0) = 0.$$

Part I: Official In-Class Quizzes (Winter 2025)

QUIZ Question 1: First-Order Nonlinear Separable IVP (Official Quiz #1-G · 15 Minutes)

- **Score Weighting:** 10 Marks
- **Problem Statement:** Solve the following equation using a method seen during the lectures. Give your answer in explicit form:

$$yy' + x^2 = 4, \quad y(0) = 2$$

1. Mathematical Classification

This is a first-order, non-linear ordinary differential equation that is separable into the form $g(y) dy = f(x) dx$.

2. Analytical Step-by-Step Derivation

1. ****Isolate the differential terms:****

Express y' as $\frac{dy}{dx}$:

$$y \frac{dy}{dx} = 4 - x^2$$

2. ****Separate the variables:****

Multiply both sides by dx :

$$y dy = (4 - x^2) dx$$

3. ****Integrate both sides:****

$$\int y dy = \int (4 - x^2) dx$$

$$\frac{y^2}{2} = 4x - \frac{x^3}{3} + C$$

4. ****Solve algebraically for y^2 :**

Multiply the entire relation by 2:

$$y^2 = 8x - \frac{2}{3}x^3 + C_1 \quad (\text{where } C_1 = 2C)$$

5. ****Apply the initial condition $y(0) = 2$:**

$$2^2 = 8(0) - \frac{2}{3}(0)^3 + C_1 \implies 4 = C_1$$

6. ****Formulate the explicit solution:****

$$y^2 = 8x - \frac{2}{3}x^3 + 4$$

Taking the square root gives $y(x) = \pm \sqrt{8x - \frac{2}{3}x^3 + 4}$. Because the initial condition requires $y(0) = +2 > 0$, we must select the positive branch:

$$y(x) = \sqrt{8x - \frac{2}{3}x^3 + 4}$$

3. Verification & Interval of Definition

* Differentiating $y(x)$:

$$y'(x) = \frac{8 - 2x^2}{2\sqrt{8x - \frac{2}{3}x^3 + 4}} = \frac{4 - x^2}{y(x)} \implies yy' = 4 - x^2 \implies yy' + x^2 = 4 \quad \checkmark$$

* The solution remains valid on the open interval containing $x = 0$ where $8x - \frac{2}{3}x^3 + 4 > 0$.

Final Explicit Answer:

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$$y(x) = \sqrt{8x - \frac{2}{3}x^3 + 4}$$

QUIZ Question 2: Nonlinear Rate of Change & Proportionality IVP (Official Quiz #2-G · 15 Minutes)

• **Score Weighting:** 10 Marks

• **Problem Statement:** Consider a process (y) as a function of time (t). This process is known to behave in the following way: The rate of change of y as a function of time is inversely proportional to the square of y .

* (a) Find the expression of the process as a function of time.

* (b) Find the value of y at $t = 5$, if we know $y(0) = 1$ and $y(1) = 2$. Show all your steps.

1. Mathematical Formulation

* The phrase "rate of change of y is inversely proportional to the square of y " translates to:

$$\frac{dy}{dt} \propto \frac{1}{y^2} \implies \frac{dy}{dt} = \frac{k}{y^2} = ky^{-2}$$

where k is a positive constant of proportionality.

2. Analytical Step-by-Step Derivation

**Part (a): General and Particular Expression $y(t)$ **

1. **Separate variables:**

$$y^2 dy = k dt$$

2. **Integrate both sides:**

$$\begin{aligned} \int y^2 dy &= \int k dt \implies \frac{y^3}{3} = kt + C \\ y^3 &= 3kt + 3C = 3kt + C_1 \quad (\text{where } C_1 = 3C) \\ y(t) &= \sqrt[3]{3kt + C_1} \end{aligned}$$

3. **Determine constant C_1 using initial condition $y(0) = 1$:

$$1 = \sqrt[3]{3k(0) + C_1} \implies 1 = \sqrt[3]{C_1} \implies C_1 = 1$$

Thus, $y(t) = \sqrt[3]{3kt + 1}$.

4. **Determine the proportionality constant k using condition $y(1) = 2$:

$$2 = \sqrt[3]{3k(1) + 1} \implies 2^3 = 3k + 1 \implies 8 = 3k + 1 \implies 3k = 7 \implies k = \frac{7}{3}$$

5. **Substitute $3k = 7$ back into the expression:

$$y(t) = \sqrt[3]{7t + 1}$$

**Part (b): Value of Process at $t = 5$ **

Substitute $t = 5$ into the derived function:

$$y(5) = \sqrt[3]{7(5) + 1} = \sqrt[3]{35 + 1} = \sqrt[3]{36} \approx 3.3019$$

Final Explicit Answers:

> (a)

$$y(t) = \sqrt[3]{7t + 1}$$

> (b)

$$y(5) = \sqrt[3]{36}$$

QUIZ **Question 3: Second-Order Nonhomogeneous ODE via Undetermined Coefficients**
(Official Quiz #3-G · 15 Minutes)

- **Score Weighting:** 10 Marks
- **Problem Statement:** Find the general solution of the equation using the Undetermined Coefficient Technique:

$$y'' + 5y' + 4y = \cos(3x)$$

1. Method Overview

The complete general solution consists of the sum of the complementary homogeneous solution $y_c(x)$ and a particular solution $y_p(x)$:

$$y(x) = y_c(x) + y_p(x)$$

2. Analytical Step-by-Step Derivation

1. **Find the Complementary Solution $y_c(x)$:**

Set the right-hand side to zero: $y'' + 5y' + 4y = 0$.

Auxiliary / Characteristic equation:

$$m^2 + 5m + 4 = 0 \iff (m + 4)(m + 1) = 0$$

The roots are distinct real numbers: $m_1 = -4$ and $m_2 = -1$.

$$y_c(x) = C_1 e^{-4x} + C_2 e^{-x}$$

2. **Formulate the Particular Trial Solution $y_p(x)$:**

The driving function is $g(x) = \cos(3x)$. Since $\pm 3i$ is not a root of the characteristic equation, no modification factor is needed.

$$y_p(x) = A \cos(3x) + B \sin(3x)$$

3. **Compute the derivatives:**

$$y_p'(x) = -3A \sin(3x) + 3B \cos(3x)$$

$$y_p''(x) = -9A \cos(3x) - 9B \sin(3x)$$

4. **Substitute into the differential operator $L[y] = y'' + 5y' + 4y$:**

$$(-9A \cos(3x) - 9B \sin(3x)) + 5(-3A \sin(3x) + 3B \cos(3x)) + 4(A \cos(3x) + B \sin(3x)) = \cos(3x)$$

Group coefficients of $\cos(3x)$ and $\sin(3x)$:

$$\cos(3x) : (-9A + 15B + 4A) = -5A + 15B = 1 \quad \text{--- [Eq. 1]}$$

$$\sin(3x) : (-9B - 15A + 4B) = -15A - 5B = 0 \quad \text{--- [Eq. 2]}$$

5. **Solve the linear algebraic system:**

From [Eq. 2]:

$$-15A = 5B \implies B = -3A$$

Substitute into [Eq. 1]:

$$-5A + 15(-3A) = 1 \implies -5A - 45A = 1 \implies -50A = 1 \implies A = -\frac{1}{50}$$

Calculate B :

$$B = -3 \left(-\frac{1}{50} \right) = \frac{3}{50}$$

Thus, the particular solution is:

$$y_p(x) = -\frac{1}{50}\cos(3x) + \frac{3}{50}\sin(3x)$$

6. **Form the full general solution:**

$$y(x) = C_1 e^{-4x} + C_2 e^{-x} - \frac{1}{50}\cos(3x) + \frac{3}{50}\sin(3x)$$

Final Explicit Answer:

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$$y(x) = C_1 e^{-4x} + C_2 e^{-x} - \frac{1}{50}\cos(3x) + \frac{3}{50}\sin(3x)$$

QUIZ **Question 4: Power Series Solution about Ordinary Point $x_0 = 0$ (Official Quiz #4 · 20 Minutes)**

- **Score Weighting:** 10 Marks
- **Problem Statement:** Using Power Series, find the solution of the equation around the ordinary point 0 (first 6 non-zero terms):

$$y'' - (x + 1)y = 0$$

Steps Required:

1. Trial solution
2. Sum the power series (adjust k in the power and then align summation indices)
3. Find the recurrence relation
4. Write the answer

1. Verification of Ordinary Point

The standard form is $y'' + P(x)y' + Q(x)y = 0$, where $P(x) = 0$ and $Q(x) = -(x + 1)$. Both coefficients are polynomials and analytic everywhere. Hence, $x_0 = 0$ is an ordinary point with an infinite radius of convergence ($R = \infty$).

2. Analytical Step-by-Step Derivation

1. **Trial Series and Derivatives:**

$$y(x) = \sum_{n=0}^{\infty} c_n x^n, \quad y'(x) = \sum_{n=1}^{\infty} n c_n x^{n-1}, \quad y''(x) = \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2}$$

2. **Substitute into $y'' - xy - y = 0$:

$$\sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} - x \sum_{n=0}^{\infty} c_n x^n - \sum_{n=0}^{\infty} c_n x^n = 0$$

Distribute x into the second summation:

$$\sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} - \sum_{n=0}^{\infty} c_n x^{n+1} - \sum_{n=0}^{\infty} c_n x^n = 0$$

3. **Shift Indices to match power x^k :

* **Term 1:** Let $k = n - 2 \implies n = k + 2$. When $n = 2$, $k = 0$:

$$\sum_{k=0}^{\infty} (k+2)(k+1) c_{k+2} x^k$$

* **Term 2:** Let $k = n + 1 \implies n = k - 1$. When $n = 0$, $k = 1$:

$$\sum_{k=1}^{\infty} c_{k-1} x^k$$

* **Term 3:** Let $k = n$. When $n = 0$, $k = 0$:

$$\sum_{k=0}^{\infty} c_k x^k$$

4. ****Align Summation Indices to $k = 1$.****

Extract the $k = 0$ terms from the first and third summations:

$$* \text{ For } k = 0: (0 + 2)(0 + 1)c_2 - c_0 = 2c_2 - c_0$$

Setting the constant term to zero gives:

$$2c_2 - c_0 = 0 \implies c_2 = \frac{c_0}{2}$$

For powers $k \geq 1$, combine the summations:

$$\sum_{k=1}^{\infty} [(k+2)(k+1)c_{k+2} - c_k - c_{k-1}]x^k = 0$$

5. ****Recurrence Relation:****

Setting the coefficient of x^k to zero for all $k \geq 1$:

$$(k+2)(k+1)c_{k+2} - c_k - c_{k-1} = 0 \implies c_{k+2} = \frac{c_k + c_{k-1}}{(k+2)(k+1)}, \quad \text{for } k \geq 1$$

6. ****Iterate to find higher-order coefficients in terms of c_0 and c_1 .****

* **$k = 1$.***

$$c_3 = \frac{c_1 + c_0}{(3)(2)} = \frac{c_0}{6} + \frac{c_1}{6}$$

* **$k = 2$.***

$$c_4 = \frac{c_2 + c_1}{(4)(3)} = \frac{\frac{c_0}{2} + c_1}{12} = \frac{c_0}{24} + \frac{c_1}{12}$$

* **$k = 3$.***

$$c_5 = \frac{c_3 + c_2}{(5)(4)} = \frac{\left(\frac{c_0}{6} + \frac{c_1}{6}\right) + \frac{c_0}{2}}{20} = \frac{\frac{4c_0}{6} + \frac{c_1}{6}}{20} = \frac{\frac{2}{3}c_0 + \frac{1}{6}c_1}{20} = \frac{c_0}{30} + \frac{c_1}{120}$$

7. ****Assemble the General Solution:****

Group terms by the two linearly independent arbitrary constants c_0 and c_1 :

$$y(x) = c_0 y_1(x) + c_1 y_2(x)$$

where:

$$y_1(x) = 1 + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \frac{1}{24}x^4 + \frac{1}{30}x^5 + \dots$$

$$y_2(x) = x + \frac{1}{6}x^3 + \frac{1}{12}x^4 + \frac{1}{120}x^5 + \dots$$

Final Explicit Answer:

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$$y(x) = c_0 \left(1 + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \frac{x^5}{30} + \dots \right) + c_1 \left(x + \frac{x^3}{6} + \frac{x^4}{12} + \frac{x^5}{120} + \dots \right)$$

Part II: Official Term Test 1 — Version 2 (Winter 2025 · Feb 21, 2025)

TERM TEST

Question 5: First-Order Separable & Linear ODEs (Test 1 · Problem #1/20)

- **Score Weighting:** 20 Marks

- **Problem Statement:** Find the general solution of the first-order ODEs below:

1. $\frac{dy}{dx} = \sin(3x)$

2. $y' + \frac{3}{x}y = x^4$

1. Solution to Sub-problem 1: $\frac{dy}{dx} = \sin(3x)$

- **Classification:** First-order directly integrable ordinary differential equation.
- **Derivation:**

$$dy = \sin(3x) dx \implies y = \int \sin(3x) dx = -\frac{1}{3}\cos(3x) + C$$

- **Boxed Result:**

$$y(x) = -\frac{1}{3}\cos(3x) + C$$

2. Solution to Sub-problem 2: $y' + \frac{3}{x}y = x^4$

- **Classification:** First-order linear ODE in standard form $y' + P(x)y = Q(x)$, with $P(x) = \frac{3}{x}$ and $Q(x) = x^4$.
- **Derivation:**

1. **Compute the Integrating Factor $I(x)$:**

$$I(x) = e^{\int P(x) dx} = e^{\int \frac{3}{x} dx} = e^{3 \ln |x|} = |x|^3$$

For $x > 0$, $I(x) = x^3$.

2. **Multiply the differential equation by $I(x)$:

$$x^3 y' + 3x^2 y = x^3 \cdot x^4 \iff \frac{d}{dx} [x^3 y] = x^7$$

3. **Integrate both sides with respect to x :

$$x^3 y = \int x^7 dx = \frac{x^8}{8} + C$$

4. **Solve explicitly for $y(x)$:

$$y(x) = \frac{x^5}{8} + Cx^{-3} = \frac{x^5}{8} + \frac{C}{x^3}$$

Final Explicit Answers:

> 1.

$$y(x) = -\frac{1}{3}\cos(3x) + C$$

> 2.

$$y(x) = \frac{x^5}{8} + \frac{C}{x^3}$$

TERM TEST**Question 6: Complex Numbers Arithmetic & Polar Modulus (Test 1 · Problem #2/15)**

- **Score Weighting:** 15 Marks

- **Problem Statement:** Consider the following two complex numbers:

$$z_1 = 4 + 5i \quad \text{and} \quad z_2 = -1 + 2i$$

- * (a) Find $z_1 + z_2$
- * (b) Find $\frac{z_1}{z_2}$ and the modulus of the ratio.
- * (c) Find $z_1 z_2$

Analytical Step-by-Step Derivation

****Part (a): Addition $z_1 + z_2$ ****

$$z_1 + z_2 = (4 + 5i) + (-1 + 2i) = (4 - 1) + (5 + 2)i = 3 + 7i$$

****Part (b): Division $\frac{z_1}{z_2}$ and Modulus $\left|\frac{z_1}{z_2}\right|$ ****

1. Multiply numerator and denominator by the complex conjugate $\bar{z}_2 = -1 - 2i$:

$$\frac{z_1}{z_2} = \frac{4 + 5i}{-1 + 2i} = \frac{(4 + 5i)(-1 - 2i)}{(-1)^2 + (2)^2} = \frac{-4 - 8i - 5i - 10i^2}{1 + 4}$$

Since $i^2 = -1$:

$$\frac{z_1}{z_2} = \frac{-4 - 13i + 10}{5} = \frac{6 - 13i}{5} = \frac{6}{5} - \frac{13}{5}i$$

2. Modulus of the quotient:

Using modulus properties $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$:

$$|z_1| = \sqrt{4^2 + 5^2} = \sqrt{16 + 25} = \sqrt{41}$$

$$|z_2| = \sqrt{(-1)^2 + 2^2} = \sqrt{1 + 4} = \sqrt{5}$$

$$\left|\frac{z_1}{z_2}\right| = \frac{\sqrt{41}}{\sqrt{5}} = \sqrt{\frac{41}{5}} = \frac{\sqrt{205}}{5} \approx 2.8636$$

Direct Cartesian Check:

$$\sqrt{\left(\frac{6}{5}\right)^2 + \left(-\frac{13}{5}\right)^2} = \sqrt{\frac{36 + 169}{25}} = \sqrt{\frac{205}{25}} = \frac{\sqrt{205}}{5} \quad \checkmark$$

****Part (c): Multiplication $z_1 z_2$ ****

$$z_1 z_2 = (4 + 5i)(-1 + 2i) = 4(-1) + 4(2i) + 5i(-1) + 5i(2i) = -4 + 8i - 5i + 10(-1) = -14 + 3i$$

Final Explicit Answers:

> (a)

$$z_1 + z_2 = 3 + 7i$$

> (b)

$$\frac{z_1}{z_2} = \frac{6}{5} - \frac{13}{5}i, \quad \left| \frac{z_1}{z_2} \right| = \sqrt{\frac{41}{5}} = \frac{\sqrt{205}}{5}$$

> (c)

$$z_1 z_2 = -14 + 3i$$

TERM TEST**Question 7: Exact Differential Equation IVP (Test 1 · Problem #3/20)**

- **Score Weighting:** 20 Marks
- **Problem Statement:** Find the particular solution of the following IVP:

$$(5y + 3t - 5) dt + (6y + 5t) dy = 0, \quad y(-1) = 0$$

1. Test for Exactness

Let $M(t, y) = 5y + 3t - 5$ and $N(t, y) = 6y + 5t$.

$$\frac{\partial M}{\partial y} = 5, \quad \frac{\partial N}{\partial t} = 5$$

Since $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial t} = 5$ everywhere on $\mathbb{R}^2$, the differential equation is **exact**.

2. Construction of the Potential Function $\Psi(t, y)$

There exists a potential function $\Psi(t, y)$ such that $\frac{\partial \Psi}{\partial t} = M$ and $\frac{\partial \Psi}{\partial y} = N$.

1. **Integrate $M(t, y)$ with respect to t :**

$$\Psi(t, y) = \int (5y + 3t - 5) dt = 5yt + \frac{3}{2}t^2 - 5t + h(y)$$

2. **Differentiate $\Psi(t, y)$ with respect to y and equate to $N(t, y)$:**

$$\frac{\partial \Psi}{\partial y} = 5t + h'(y) = 6y + 5t \implies h'(y) = 6y$$

3. **Integrate $h'(y)$:**

$$h(y) = 3y^2$$

4. **General Implicit Solution:**

$$3y^2 + 5ty + \frac{3}{2}t^2 - 5t = C$$

3. Apply the Initial Condition $y(-1) = 0$

Substitute $t = -1$ and $y = 0$:

$$3(0)^2 + 5(-1)(0) + \frac{3}{2}(-1)^2 - 5(-1) = C \implies 0 + 0 + \frac{3}{2} + 5 = C \implies C = \frac{13}{2}$$

Multiply through by 2 to clear fractions:

$$6y^2 + 10ty + 3t^2 - 10t - 13 = 0$$

4. Explicit Formulation

Apply the quadratic formula to solve for y :

$$y(t) = \frac{-10t \pm \sqrt{(10t)^2 - 4(6)(3t^2 - 10t - 13)}}{2(6)} = \frac{-10t \pm \sqrt{100t^2 - 72t^2 + 240t + 312}}{12}$$

$$y(t) = \frac{-10t \pm \sqrt{28t^2 + 240t + 312}}{12} = \frac{-5t \pm \sqrt{7t^2 + 60t + 78}}{6}$$

To satisfy $y(-1) = 0$:

$$\frac{-5(-1) \pm \sqrt{7(-1)^2 + 60(-1) + 78}}{6} = \frac{5 \pm \sqrt{7 - 60 + 78}}{6} = \frac{5 \pm \sqrt{25}}{6} = \frac{5 \pm 5}{6}$$

Choosing the negative sign yields $\frac{5-5}{6} = 0$.

Final Explicit Answer:

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$$y(t) = \frac{-5t - \sqrt{7t^2 + 60t + 78}}{6} \quad \text{or implicitly: } 6y^2 + 10ty + 3t^2 - 10t = 13$$

TERM TEST**Question 8: Logistic Differential Equation for Technology Adoption (Test 1 · Problem #4/25)**

- **Score Weighting:** 25 Marks

- **Problem Statement:** The number $N(t)$ of stores in Canada using a computerized checkout system is described by the initial value problem:

$$\frac{dN}{dt} = N(1 - 0.0004N), \quad N(0) = 2$$

Solve the initial value problem (show all your steps) to get $N(t)$, and find how many stores are expected to use this system for $t = 12$.

1. Identification of Parameters

This is the Verhulst logistic growth model:

$$\frac{dN}{dt} = rN \left(1 - \frac{N}{K} \right)$$

Here, intrinsic growth rate $r = 1$, and $0.0004 = \frac{1}{2500}$, meaning the carrying capacity (maximum market saturation) is $K = 2500$ stores.

2. Analytical Step-by-Step Derivation

1. ****Separate variables:****

$$\frac{dN}{N(1 - 0.0004N)} = dt \iff \frac{2500 dN}{N(2500 - N)} = dt$$

2. ****Partial Fraction Decomposition:****

$$\frac{2500}{N(2500 - N)} = \frac{A}{N} + \frac{B}{2500 - N}$$

$$2500 = A(2500 - N) + BN$$

Setting $N = 0 \implies A = 1$. Setting $N = 2500 \implies B = 1$.

$$\left(\frac{1}{N} + \frac{1}{2500 - N} \right) dN = dt$$

3. ****Integrate both sides:****

$$\ln |N| - \ln |2500 - N| = t + C \implies \ln \left| \frac{N}{2500 - N} \right| = t + C$$

$$\frac{N}{2500 - N} = C_0 e^t \quad (\text{where } C_0 = \pm e^C)$$

4. ****Determine C_0 using $N(0) = 2$:****

$$\frac{2}{2500 - 2} = C_0 e^0 \implies C_0 = \frac{2}{2498} = \frac{1}{1249}$$

5. ****Solve explicitly for $N(t)$:****

$$\frac{N}{2500 - N} = \frac{e^t}{1249} \implies 1249N = 2500e^t - Ne^t \implies N(1249 + e^t) = 2500e^t$$

$$N(t) = \frac{2500e^t}{1249 + e^t} = \frac{2500}{1 + 1249e^{-t}}$$

3. Evaluation at $t = 12$

Substitute $t = 12$:

$$N(12) = \frac{2500}{1 + 1249e^{-12}}$$

Compute e^{-12} :

$$e^{-12} \approx 6.14421 \times 10^{-6}$$

$$1249 \times e^{-12} \approx 1249 \times 6.14421 \times 10^{-6} \approx 0.007674$$

$$N(12) = \frac{2500}{1 + 0.007674} = \frac{2500}{1.007674} \approx 2480.97 \approx 2481 \text{ stores}$$

Final Explicit Answers:

>

$$N(t) = \frac{2500}{1 + 1249e^{-t}}$$

> At $t = 12$:

$$N(12) \approx 2481 \text{ stores (99.24\% of market saturation)}$$

TERM TEST Question 9: Forensic Newton's Law of Cooling (Test 1 · Problem #5/20)

- **Score Weighting:** 20 Marks

- **Problem Statement:** A dead body was found in a closed room having a constant temperature of 20°C . At the time of discovery, the body temperature was 27°C . One hour later, it was at 24°C . Assume the time of death is $t = 0$ and that a living body has a temperature of 37°C . How many hours elapsed before the body was found?

(Hint: use Newton's law of cooling: $\frac{dT}{dt} = k(T - T_m)$)

1. Mathematical Formulation

* Ambient temperature: $T_m = 20^{\circ}\text{C}$.

* Initial body temperature at death ($t = 0$): $T(0) = 37^{\circ}\text{C}$.

* Discovery occurs at an unknown elapsed time t_d . Measured: $T(t_d) = 27^{\circ}\text{C}$.

* One hour post-discovery ($t = t_d + 1$): $T(t_d + 1) = 24^{\circ}\text{C}$.

2. Analytical Step-by-Step Derivation

1. **Solve Newton's Law of Cooling:**

$$\frac{dT}{dt} = k(T - 20) \implies \frac{dT}{T - 20} = k dt$$

$$\ln |T - 20| = kt + C \implies T(t) - 20 = C_0 e^{kt} \implies T(t) = 20 + C_0 e^{kt}$$

2. **Apply Initial Condition $T(0) = 37$:

$$37 = 20 + C_0 e^0 \implies C_0 = 17$$

Therefore, the temperature trajectory is:

$$T(t) = 20 + 17e^{kt}$$

3. **Formulate the system of two observation equations:**

* At discovery $t = t_d$:

$$27 = 20 + 17e^{kt_d} \implies 17e^{kt_d} = 7 \implies e^{kt_d} = \frac{7}{17} \quad \text{--- [Eq. A]}$$

* One hour later $t = t_d + 1$:

$$24 = 20 + 17e^{k(t_d+1)} \implies 17e^{kt_d}e^k = 4 \quad \text{--- [Eq. B]}$$

4. **Determine the cooling constant k :

Substitute [Eq. A] into [Eq. B]:

$$17 \left(\frac{7}{17} \right) e^k = 4 \implies 7e^k = 4 \implies e^k = \frac{4}{7}$$

$$k = \ln \left(\frac{4}{7} \right) \approx -0.559616 \text{ hr}^{-1}$$

5. **Calculate the elapsed time t_d :**

From [Eq. A], take the natural logarithm:

$$kt_d = \ln\left(\frac{7}{17}\right) \implies t_d = \frac{\ln(7/17)}{k} = \frac{\ln(7/17)}{\ln(4/7)}$$

Evaluate numerical values:

$$\ln\left(\frac{7}{17}\right) = \ln(0.411765) \approx -0.887303$$

$$\ln\left(\frac{4}{7}\right) = \ln(0.571429) \approx -0.559616$$

$$t_d = \frac{-0.887303}{-0.559616} \approx 1.58555 \text{ hours}$$

Convert to hours and minutes:

$$0.58555 \times 60 \approx 35.13 \text{ minutes} \implies 1 \text{ hour, } 35 \text{ minutes and } 8 \text{ seconds}$$

Final Explicit Answer:

>

$$t_d = \frac{\ln(7/17)}{\ln(4/7)} \approx 1.59 \text{ hours (approximately 1 hour and 35 minutes)}$$

Part III: Official Term Test 2 (Winter 2025 · March 27, 2025)

TERM TEST Question 10: Cauchy-Euler Equidimensional ODE (Test 2 · Problem #1/20)

- **Score Weighting:** 20 Marks
- **Problem Statement:** Find the general solution of the ODE below:

$$x^2 y'' + 5xy' + 8y = 0 \quad (x > 0)$$

1. Mathematical Classification

This is a second-order homogeneous Cauchy-Euler (equidimensional) differential equation with constant coefficients $a = 1$, $b = 5$, and $c = 8$.

2. Analytical Step-by-Step Derivation

1. ****Trial Substitution:****

Assume a solution of the form $y(x) = x^m$.

$$y' = mx^{m-1}, \quad y'' = m(m-1)x^{m-2}$$

2. ****Substitute into the differential equation:****

$$\begin{aligned} x^2[m(m-1)x^{m-2}] + 5x[mx^{m-1}] + 8[x^m] &= 0 \\ x^m[m(m-1) + 5m + 8] &= 0 \end{aligned}$$

3. ****Form and solve the auxiliary characteristic equation:****

$$m^2 - m + 5m + 8 = 0 \iff m^2 + 4m + 8 = 0$$

Use the quadratic formula:

$$m = \frac{-4 \pm \sqrt{4^2 - 4(1)(8)}}{2} = \frac{-4 \pm \sqrt{16 - 32}}{2} = \frac{-4 \pm \sqrt{-16}}{2} = -2 \pm 2i$$

The roots are complex conjugates: $\alpha = -2$ and $\beta = 2$.

4. ****General Solution Formula for Complex Conjugate Roots:****

For Cauchy-Euler equations with roots $\alpha \pm i\beta$:

$$y(x) = x^\alpha [C_1 \cos(\beta \ln x) + C_2 \sin(\beta \ln x)]$$

Substitute $\alpha = -2$ and $\beta = 2$:

$$y(x) = x^{-2} [C_1 \cos(2 \ln x) + C_2 \sin(2 \ln x)] = \frac{C_1 \cos(2 \ln x) + C_2 \sin(2 \ln x)}{x^2}$$

Final Explicit Answer:

>

$$y(x) = x^{-2} [C_1 \cos(2 \ln x) + C_2 \sin(2 \ln x)]$$

TERM TEST**Question 11: Variation of Parameters for 2nd-Order ODE (Test 2 · Problem #2/20)**

- **Score Weighting:** 20 Marks
- **Problem Statement:** Using Variation of parameter technique, solve:

$$y'' - 4y' + 4y = (x + 1)e^{2x}$$

1. Homogeneous Solution and Basis Functions

Auxiliary equation:

$$m^2 - 4m + 4 = 0 \iff (m - 2)^2 = 0 \implies m_1 = m_2 = 2 \text{ (repeated real root)}$$

Fundamental set of solutions:

$$y_1(x) = e^{2x}, \quad y_2(x) = xe^{2x}$$

$$y_c(x) = C_1e^{2x} + C_2xe^{2x}$$

2. Wronskian Determinant Calculation

$$W(y_1, y_2)(x) = \begin{vmatrix} e^{2x} & xe^{2x} \\ 2e^{2x} & (1+2x)e^{2x} \end{vmatrix} = e^{2x}(1+2x)e^{2x} - 2xe^{2x}e^{2x} = e^{4x}(1+2x-2x) = e^{4x} \neq 0$$

3. Variation of Parameters Formula

With driving term $g(x) = (x + 1)e^{2x}$, particular solution is $y_p(x) = u_1(x)y_1(x) + u_2(x)y_2(x)$:

1. **Calculate $u_1(x)$:**

$$u_1'(x) = -\frac{y_2(x)g(x)}{W(x)} = -\frac{(xe^{2x})(x+1)e^{2x}}{e^{4x}} = -\frac{x(x+1)e^{4x}}{e^{4x}} = -(x^2 + x)$$

$$u_1(x) = \int -(x^2 + x) dx = -\frac{x^3}{3} - \frac{x^2}{2}$$

2. **Calculate $u_2(x)$:**

$$u_2'(x) = \frac{y_1(x)g(x)}{W(x)} = \frac{(e^{2x})(x+1)e^{2x}}{e^{4x}} = \frac{(x+1)e^{4x}}{e^{4x}} = x + 1$$

$$u_2(x) = \int (x + 1) dx = \frac{x^2}{2} + x$$

3. **Assemble and Simplify $y_p(x)$:**

$$y_p(x) = \left(-\frac{x^3}{3} - \frac{x^2}{2}\right)e^{2x} + \left(\frac{x^2}{2} + x\right)xe^{2x} = e^{2x} \left(-\frac{x^3}{3} - \frac{x^2}{2} + \frac{x^3}{2} + x^2\right)$$

Combine like terms:

$$-\frac{1}{3}x^3 + \frac{1}{2}x^3 = \frac{1}{6}x^3, \quad -\frac{1}{2}x^2 + x^2 = \frac{1}{2}x^2$$

$$y_p(x) = e^{2x} \left(\frac{x^3}{6} + \frac{x^2}{2}\right)$$

4. **General Solution:**

$$y(x) = C_1 e^{2x} + C_2 x e^{2x} + e^{2x} \left(\frac{x^3}{6} + \frac{x^2}{2} \right)$$

Final Explicit Answer:

>

$$y(x) = C_1 e^{2x} + C_2 x e^{2x} + e^{2x} \left(\frac{x^3}{6} + \frac{x^2}{2} \right)$$

TERM TEST**Question 12: Second-Order Homogeneous IVP with Repeated Roots (Test 2 · Problem #3/20)**

- **Score Weighting:** 20 Marks
- **Problem Statement:** Find the particular solution of the following IVP:

$$y'' + y' + 0.25y = 0, \quad y(0) = 3, \quad y'(0) = -3.5$$

1. Analytical Step-by-Step Derivation

1. **Auxiliary Equation:**

$$m^2 + m + 0.25 = 0 \iff m^2 + m + \frac{1}{4} = 0 \iff \left(m + \frac{1}{2}\right)^2 = 0$$

Repeated real root: $m_1 = m_2 = -0.5 = -\frac{1}{2}$.

2. **General Solution:**

$$y(x) = (C_1 + C_2x)e^{-0.5x}$$

3. **Compute the Derivative $y'(x)$:**

$$y'(x) = C_2e^{-0.5x} - 0.5(C_1 + C_2x)e^{-0.5x} = (C_2 - 0.5C_1 - 0.5C_2x)e^{-0.5x}$$

4. **Apply Initial Conditions:**

* $y(0) = 3$:

$$(C_1 + C_2(0))e^0 = 3 \implies C_1 = 3$$

* $y'(0) = -3.5$:

$$C_2 - 0.5(3) = -3.5 \implies C_2 - 1.5 = -3.5 \implies C_2 = -2$$

5. **Formulate the Particular Solution:**

$$y(x) = (3 - 2x)e^{-0.5x}$$

Final Explicit Answer:

>

$$y(x) = (3 - 2x)e^{-0.5x}$$

TERM TEST**Question 13: Mechanical Vibrations of Damped Spring-Mass System (Test 2 · Problem #4/20)**

- **Score Weighting:** 20 Marks

- **Problem Statement:** A 2 kg mass is attached to a spring of constant 72 N/m. The system is horizontal and the ground provides a damping force equal to 4 times the velocity.

- * (a) Write the equation for the forces involved, starting with Newton's second law (sum of forces).
- * (b) Solve the equation found in (a).
- * (c) Find the velocity of the mass 1 s after release if the mass is released from a position $x(0) = 2$ m (released from rest, $x'(0) = 0$).

Analytical Step-by-Step Derivation

****Part (a): Newton's Second Law Force Balance****

1. System parameters:

* Mass: $m = 2$ kg

* Damping coefficient: $c = 4$ N · s/m (damping force opposes velocity: $F_d = -c \frac{dx}{dt}$)

* Spring constant: $k = 72$ N/m (restoring force opposes displacement: $F_s = -kx$)

2. Sum of forces:

$$\sum F = ma \implies -kx - c \frac{dx}{dt} = m \frac{d^2x}{dt^2}$$

$$mx'' + cx' + kx = 0 \implies 2x'' + 4x' + 72x = 0$$

Dividing by 2 gives the standard monic form:

$$x'' + 2x' + 36x = 0$$

****Part (b): Solution of the Homogeneous ODE****

1. Characteristic equation:

$$m^2 + 2m + 36 = 0 \implies m = \frac{-2 \pm \sqrt{2^2 - 4(1)(36)}}{2} = \frac{-2 \pm \sqrt{4 - 144}}{2} = \frac{-2 \pm \sqrt{-140}}{2} = -1 \pm i\sqrt{35}$$

2. Underdamped motion with decay factor $\alpha = 1$ and damped natural frequency

$$\omega_d = \sqrt{35} \approx 5.9161 \text{ rad/s:}$$

$$x(t) = e^{-t} \left[C_1 \cos(\sqrt{35}t) + C_2 \sin(\sqrt{35}t) \right]$$

****Part (c): Determination of Velocity at $t = 1$ s****

1. Apply initial conditions $x(0) = 2$ m and $x'(0) = 0$:

$$x(0) = C_1 = 2$$

2. Differentiate $x(t)$ using the product rule:

$$x'(t) = -e^{-t}[C_1 \cos(\sqrt{35}t) + C_2 \sin(\sqrt{35}t)] + e^{-t}[-\sqrt{35}C_1 \sin(\sqrt{35}t) + \sqrt{35}C_2 \cos(\sqrt{35}t)]$$

$$x'(t) = e^{-t}[(\sqrt{35}C_2 - C_1) \cos(\sqrt{35}t) - (C_2 + \sqrt{35}C_1) \sin(\sqrt{35}t)]$$

3. Apply $x'(0) = 0$:

$$\sqrt{35}C_2 - C_1 = 0 \implies \sqrt{35}C_2 - 2 = 0 \implies C_2 = \frac{2}{\sqrt{35}}$$

4. Simplified velocity function $v(t) = x'(t)$:

Since $\sqrt{35}C_2 - C_1 = 0$, the cosine component vanishes completely:

$$v(t) = x'(t) = -e^{-t}(C_2 + \sqrt{35}C_1) \sin(\sqrt{35}t)$$

Substitute $C_1 = 2$ and $C_2 = \frac{2}{\sqrt{35}}$:

$$C_2 + \sqrt{35}C_1 = \frac{2}{\sqrt{35}} + 2\sqrt{35} = \frac{2 + 70}{\sqrt{35}} = \frac{72}{\sqrt{35}}$$

$$v(t) = -\frac{72}{\sqrt{35}}e^{-t} \sin(\sqrt{35}t)$$

5. Evaluate at $t = 1$ s:

$$v(1) = -\frac{72}{\sqrt{35}}e^{-1} \sin(\sqrt{35})$$

Numerical values:

$$\sqrt{35} \approx 5.91608 \text{ rad}$$

$$\sin(5.91608 \text{ rad}) \approx -0.35825$$

$$e^{-1} \approx 0.367879$$

$$\frac{72}{\sqrt{35}} \approx \frac{72}{5.91608} \approx 12.1699$$

$$v(1) \approx -(12.1699) \times (0.367879) \times (-0.35825) \approx +1.604 \text{ m/s}$$

Final Explicit Answers:

> (a)

$$2x'' + 4x' + 72x = 0 \iff x'' + 2x' + 36x = 0$$

> (b)

$$x(t) = e^{-t} [C_1 \cos(\sqrt{35}t) + C_2 \sin(\sqrt{35}t)]$$

> (c)

$$v(1) = -\frac{72}{\sqrt{35}}e^{-1} \sin(\sqrt{35}) \approx +1.60 \text{ m/s}$$

TERM TEST**Question 14: Forced Harmonic Oscillator with Damping IVP (Test 2 · Problem #5/20)**

- **Score Weighting:** 20 Marks
- **Problem Statement:** A force is driving a system of mass and spring such as:

$$x'' + 4x' + 4x = 10 \cos(3t)$$

Find the equation of motion if the initial position and velocity are 0 ($x(0) = 0, x'(0) = 0$).

Analytical Step-by-Step Derivation

1. ****Homogeneous Complementary Solution $x_h(t)$ ****

Characteristic equation:

$$m^2 + 4m + 4 = 0 \iff (m + 2)^2 = 0 \implies m_1 = m_2 = -2 \text{ (critically damped)}$$

$$x_h(t) = (C_1 + C_2 t)e^{-2t}$$

2. ****Particular Solution $x_p(t)$ via Undetermined Coefficients****

Driving frequency is $\omega = 3$. Assume:

$$x_p(t) = A \cos(3t) + B \sin(3t)$$

Compute derivatives:

$$x_p'(t) = -3A \sin(3t) + 3B \cos(3t)$$

$$x_p''(t) = -9A \cos(3t) - 9B \sin(3t)$$

Substitute into $x'' + 4x' + 4x = 10 \cos(3t)$:

$$(-9A \cos(3t) - 9B \sin(3t)) + 4(-3A \sin(3t) + 3B \cos(3t)) + 4(A \cos(3t) + B \sin(3t)) = 10 \cos(3t)$$

Collect terms for $\cos(3t)$ and $\sin(3t)$:

$$\cos(3t) : (-9A + 12B + 4A) = -5A + 12B = 10 \quad \text{--- [Eq. 1]}$$

$$\sin(3t) : (-9B - 12A + 4B) = -12A - 5B = 0 \implies B = -\frac{12}{5}A \quad \text{--- [Eq. 2]}$$

3. ****Solve for A and B ****

Substitute [Eq. 2] into [Eq. 1]:

$$-5A + 12 \left(-\frac{12}{5}A \right) = 10 \implies -5A - \frac{144}{5}A = 10 \implies -\frac{169}{5}A = 10$$

$$A = -\frac{50}{169}$$

Compute B :

$$B = -\frac{12}{5} \left(-\frac{50}{169} \right) = \frac{120}{169}$$

Therefore:

$$x_p(t) = -\frac{50}{169}\cos(3t) + \frac{120}{169}\sin(3t)$$

4. **General Solution:**

$$x(t) = (C_1 + C_2t)e^{-2t} - \frac{50}{169}\cos(3t) + \frac{120}{169}\sin(3t)$$

5. **Apply Initial Conditions $x(0) = 0$ and $x'(0) = 0$:

* $x(0) = 0$:

$$(C_1 + 0)e^0 - \frac{50}{169}(1) + \frac{120}{169}(0) = 0 \implies C_1 - \frac{50}{169} = 0 \implies C_1 = \frac{50}{169}$$

* Differentiate $x(t)$:

$$x'(t) = C_2e^{-2t} - 2(C_1 + C_2t)e^{-2t} + \frac{150}{169}\sin(3t) + \frac{360}{169}\cos(3t)$$

* $x'(0) = 0$:

$$C_2 - 2C_1 + \frac{360}{169} = 0 \implies C_2 - 2\left(\frac{50}{169}\right) + \frac{360}{169} = 0$$

$$C_2 - \frac{100}{169} + \frac{360}{169} = 0 \implies C_2 + \frac{260}{169} = 0 \implies C_2 = -\frac{260}{169} = -\frac{20}{13}$$

6. **Final Equation of Motion:**

$$x(t) = \left(\frac{50}{169} - \frac{260}{169}t\right)e^{-2t} - \frac{50}{169}\cos(3t) + \frac{120}{169}\sin(3t)$$

Final Explicit Answer:

>

$$x(t) = \left(\frac{50}{169} - \frac{260}{169}t\right)e^{-2t} - \frac{50}{169}\cos(3t) + \frac{120}{169}\sin(3t)$$

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